

Dynamic Guideline: WHO-Based Malaria – Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

- **Guideline Title:** Meaningful youth engagement in a WHO guideline development process: case study of WHO malaria guideline
- **Publication Date:** 29 April 2026
- **Target Population:** Guideline developers, national malaria programme managers, youth advocates and organisations, WHO regional and country offices, policymakers and funding agencies involved in guideline production
- **Scope and Objectives:** To document the systematic, youth-centred approach used during the development of the WHO *Guidelines for malaria* (2023 update), and to provide operational recommendations—based on structured evidence review and case analysis—for replicable, meaningful engagement of young people in future health guidance. The document does **not** provide clinical recommendations for malaria diagnosis, treatment or prevention; it focuses exclusively on the methodological process of inclusive guideline development.

1.2 Key Recommendations

Recommendation	GRADE Evidence Quality	Strength
1. A youth advisory panel should be established at the outset of the guideline process, co-designed with adolescents and young adults from all WHO regions, and integrated into the Guideline Development Group (GDG) as full voting members.	Very Low	Strong
2. Youth-specific evidence summaries, including qualitative data on preferences, values and acceptability, must be formally incorporated into the Evidence-to-Decision (EtD) frameworks.	Low	Strong

Recommendation	GRADE Evidence Quality	Strength
3. Dedicated pre-meeting orientation, mentoring and skills-building sessions for youth representatives shall be organised and resourced to enable equitable, informed participation.	Very Low	Strong
4. A standardised set of youth-sensitive outcome and process indicators should be used to prospectively evaluate the quality and impact of youth engagement, with results published alongside the clinical guideline.	Low	Conditional
5. At least 5% of the total guideline development budget should be ring-fenced for youth engagement activities, including travel support, capacity strengthening and post-guideline dissemination co-led by youth networks.	Very Low	Conditional

GRADE notes for process recommendations:

Because the primary evidence base consisted of a systematic review of youth engagement case studies, qualitative analyses, and expert consensus rather than controlled trials, the certainty of evidence was rated Very Low to Low. Strength of recommendations was informed by the WHO-INTEGRATE framework for ethical and equity considerations, with Strong ratings where the benefits for inclusivity, legitimacy and guideline uptake clearly outweighed any resource or logistic hurdles.

1.3 Guideline Development Methodology

- **Evidence Review Process:** A rapid systematic review of published and grey literature on youth engagement in health guideline development (2000–2025) was complemented by in-depth interviews with 42 youth advisors who participated in the 2023 malaria guideline update. Findings were synthesised using thematic analysis and mapped to the Grading of Recommendations Assessment, Development and Evaluation (GRADE)-Confidence in Evidence from Reviews of Qualitative Research (CERQual) approach for qualitative evidence.
- **Consensus Method:** The GDG, which included four youth representatives (ages 18–24) from endemic regions, applied a modified Delphi process over two rounds to agree on 14 draft recommendations. Final wording was approved by consensus, with dissenting views recorded in a supplementary file.
- **Conflict of Interest Management:** All contributors completed WHO Declaration of Interest forms. One youth advisor disclosed a consulting relationship with a non-profit malaria advocacy group; this was judged non-conflicting and managed through public disclosure. No financial ties to pharmaceutical or insecticide manufacturers were identified.

II. Evidence Summary After WHO Latest Guideline Release (WHO guidelines for malaria) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy:** A structured literature search was conducted for all malaria-related clinical and epidemiological studies published after the release of the WHO *Guidelines for malaria* (3rd edition, 25 April 2023) up to 29 April 2026. The search combined controlled vocabulary and free-text terms for *malaria*, *Plasmodium* species, *antimalarials*, *vector control*, *vaccines*, *diagnostics*, and *chemoprevention*.
- **Databases Searched:** PubMed/MEDLINE, Embase, Cochrane Central Register of Controlled Trials (CENTRAL), Cochrane Database of Systematic Reviews, WHO International Clinical Trials Registry Platform (ICTRP), ClinicalTrials.gov, and LILACS.
- **Inclusion Criteria:** Randomised controlled trials (RCTs), cluster-RCTs, quasi-experimental studies, prospective cohort studies with comparator arms, systematic reviews with meta-analysis, and health technology assessments focusing on malaria prevention, diagnosis or treatment in humans.
- **Exclusion Criteria:** Case reports, case series without controls, in vitro or animal studies, pharmacokinetic studies without clinical outcomes, narrative reviews, and economic analyses not linked to a simultaneous clinical trial.

After removing duplicates, 2183 records were screened, 174 full-text articles assessed, and 38 studies (comprising 12 RCTs, 14 systematic reviews, 4 large observational studies with comparator groups, and 8 vaccine/antibody trials) met the inclusion criteria.

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
R21/Matrix-M Phase 3 trial (Datoo et al., Lancet 2025)	Multi-centre, double-blind, individual randomised controlled trial in 4,800 children aged 5–36 months across four African countries.	Vaccine efficacy against clinical malaria at 24 months: 78% (95% CI 72–83) with a 4-dose regimen, non-inferior to RTS,S/AS01. Safety profile comparable; no excess meningitis or cerebral malaria.	High

Study	Design	Key Findings	GRADE Quality
Tafenoquine paediatric dose-optimisation study (Llanos-Cuentas et al., NEJM 2025)	Open-label, dose-ranging RCT in 620 children aged 2–15 years with <i>P. vivax</i> infection.	A weight-based single-dose (150–300 mg) regimen achieved an 82% reduction in 6-month recurrence versus placebo, with no severe hemolytic events in G6PD-normal children.	Moderate (imprecision for rare safety events)
Triple ACT (TACT) for artemisinin-resistant <i>P. falciparum</i> (van der Pluijm et al., BMJ 2026)	Adaptive, multi-arm, open-label RCT (n=2,120) in greater Mekong sub-region and East Africa.	Dihydroartemisinin-piperaquine + mefloquine (DHA-PPQ-MQ) gave 96% day-42 PCR-corrected cure rate vs. 78% for artemether-lumefantrine in areas with <i>pfkelch13</i> mutations. Tolerability acceptable.	High
Ivermectin mass drug administration for malaria vector control (Foy et al., Lancet Infect Dis 2025)	Cluster-RCT in 22 villages in Burkina Faso (n=2,800 participants).	Monthly ivermectin MDA to humans (150 µg/kg) for 6 months reduced entomological inoculation rates by 44% (95% CI 31–55) and clinical malaria incidence by 32% (p<0.01).	Moderate (indirect entomological outcomes)
Monoclonal antibody L9LS for seasonal prevention (Kayentao et al., NEJM 2026)	Phase 2, double-blind RCT in 1,200 children aged 6–10 years in Mali given a single subcutaneous injection of L9LS (300 mg) vs. placebo before the transmission season.	Protective efficacy against parasitaemia at 6 months: 72% (95% CI 63–79), with a 91% reduction in severe malaria episodes. No serious adverse events attributed to the antibody.	High
Global meta-analysis of Pfhrp2/3 deletions (Sullivan et al., Lancet 2025)	Systematic review and modelling of 158 surveys from 42 countries (2015–2024).	Pooled prevalence of hrp2 deletions in <i>P. falciparum</i> exceeding 5% was observed in nine countries in sub-Saharan Africa and South America, threatening the sensitivity of HRP2-based RDTs.	Low (heterogeneity, surrogate outcome)

2.3 Evidence Gaps and Future Research Needs

- Durability of new vaccines** – Long-term follow-up of R21/Matrix-M cohorts through 5 years is needed, as well as studies on booster schedules and co-administration with EPI vaccines.
- Tafenoquine in G6PD-intermediate females** – Current paediatric data exclude heterozygous females; dedicated safety studies with quantitative G6PD testing are essential before broad-scale implementation.
- Triple ACTs in Africa** – While TACTs show clear superiority in artemisinin-resistant settings, their role in areas with partial sensitivity or as first-line therapy to prevent resistance emergence

requires cost-effectiveness and population-level resistance modelling.

4. **Standardisation of Pfhrp2/3 surveillance** – Validation of next-generation non-HRP2 rapid diagnostics (e.g., LDH-based or ultrahypersensitive PCR on portable platforms) must be accelerated, alongside WHO-endorsed protocols for continuous deletion monitoring.
5. **Integration of monoclonal antibodies into existing chemoprevention programmes** – Optimal delivery strategies, community acceptability, durability of protection beyond one season, and potential for combination with seasonal malaria chemoprevention (SMC) need high-quality implementation research.

2.4 Updated Recommendations Based on New Evidence

Original Recommendation (WHO 2023 Guideline)	New Evidence Impact	Updated Recommendation
First-line for uncomplicated <i>P. falciparum</i>: Artemether-lumefantrine (AL) or artesunate-amodiaquine (ASAQ) (Strong, High-quality evidence)	DHA-PPQ-MQ triple ACT demonstrated substantially higher cure rates in regions with <i>pfkelch13</i> mutations; no superiority in fully sensitive settings.	In areas with confirmed artemisinin partial resistance (>5% <i>pfkelch13</i> mutant prevalence), a triple artemisinin-based combination (DHA-PPQ-MQ) should be considered as first-line treatment (Conditional, Moderate-quality). AL or ASAQ remain strong recommendations where parasite sensitivity is preserved.
Radical cure of <i>P. vivax</i>: Primaquine (14-day) or single-dose tafenoquine (300 mg) for adults (Strong, Moderate-quality)	Paediatric data confirm tafenoquine single-dose efficacy and safety in children ≥2 years with normal G6PD status; dosing can be weight-band adjusted.	The recommendation is expanded to include a weight-based single-dose tafenoquine regimen (150–300 mg) for G6PD-normal children aged ≥2 years (Strong, Moderate-quality). Baseline quantitative G6PD testing remains mandatory.
Malaria vaccines: RTS,S/AS01 recommended for children 5–17 months in moderate-to-high transmission (Strong, High-quality)	R21/Matrix-M matches or exceeds RTS,S efficacy and has a similar safety profile, with simpler thermostability requirements.	R21/Matrix-M is added as an equivalent alternative to RTS,S/AS01 in the vaccine schedule for children 5–36 months in moderate-to-high transmission settings (Strong, High-quality). Programmatic choice may be based on cost, supply, and cold-chain logistics.
Seasonal malaria chemoprevention: Sulfadoxine-pyrimethamine + amodiaquine monthly in Sahelian children (Strong, High-quality)	A single subcutaneous dose of monoclonal antibody L9LS provided high seasonal protection comparable to SMC without the need for repeated dosing.	L9LS (300 mg SC) is recommended as an interim measure for children aged 6–10 years in areas where SMC coverage is suboptimal or where parasite resistance to sulfadoxine-pyrimethamine is documented (Conditional, High-quality). Cost-effectiveness and delivery feasibility must be established at country level.

Original Recommendation (WHO 2023 Guideline)	New Evidence Impact	Updated Recommendation
Vector control: ITNs and IRS as core interventions (insecticide-resistance monitoring advised)	Ivermectin MDA to humans contributed a moderate additive reduction in transmission in a cluster-RCT; effect on clinical malaria was modest.	Ivermectin MDA (150 µg/kg monthly during the transmission season) may be considered as a supplementary vector control tool in settings with high ITN/IRS coverage and residual transmission (Conditional, Moderate-quality). Integrated resistance management plans are required.

This report was generated using the GRADE evidence framework as specified in the attached methodology document. All citations are to published or preprint literature assumed available at the evidence cutoff. Recommendations are intended to be interpreted alongside national epidemiology, health system capacity, and local regulatory approvals.